

GAN applied to Wave Function Collapse for procedural map generation

Cédric Chopin
ESGI
Paris, France
cedricamond@hotmail.fr

Etienne Gibiat
ESGI
Paris, France
gibiatetienne@gmail.com

Nicolas Ruche
ESGI
Paris, France
nicolas@ruche.me

ABSTRACT

This paper describes the use of Generative Adversarial Network (GAN) applied to the Wave Function Collapse (WFC) algorithm for procedural content generation. The goal of this system is to enable level designers to generate coherent 3D worlds with brand new meshes generated by the GAN.

CCS CONCEPTS

• Computing methodologies → Mesh models.

KEYWORDS

Generative Adversarial Network, GAN, neural networks, Wave Function Collapse, WFC, procedural generation

ACM Reference Format:

Cédric Chopin, Etienne Gibiat, and Nicolas Ruche. 2022. GAN applied to Wave Function Collapse for procedural map generation. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH '22 Posters)*, August 07-11, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532719.3543198>

1 INTRODUCTION

Procedural Content Generation (PCG) are systems which take in certain patterns as input, and output new assets from these patterns. It moves the part of artistic work-flow from artists to programmers and speeds up asset creation. One of the drawback is the loss of control over the end result. Procedural map generation poses an important role amongst the uses of PCG. Wave Function Collapse is greedy PCG algorithm that is able to generate large consistent output from a small number of constraints. The algorithm can be used to generate procedural data in a wide variety of domains, ranging from 3D world generation to poetry generation. It quickly became popular, due to its modularity and expansibility. In the normal workflow, models are created before-hand and each model needs to be associated with meta-data indicating its constraints. An example of constraint is "A bridge must be connected to a river". The proposed system is an improvement of the WFC based on a GAN, which is a class of machine learning frameworks used, given a training set, to generate new data with similarities to the training set. The WFC works as usual but each time it needs a model, it calls

the GAN to generate it, feeding it the constraints that the mesh needs to have. The presented method was implemented in 2D but can be upgraded to 3D, it just needs more constraints and training for the GAN.

2 ALGORITHM

2.1 WFC

Each mesh has four constraints : one for each direction in the 2D space.

The WFC requires a grid in which the meshes will be put in a coherent way. The first mesh to be put is chosen randomly. Then, when the WFC wants to put adjacent meshes, they must satisfy the constraints of the initial mesh. Let's say the first mesh was a straight path, then we can't put a wall at its end, we must choose for example a mesh which is a dead end.

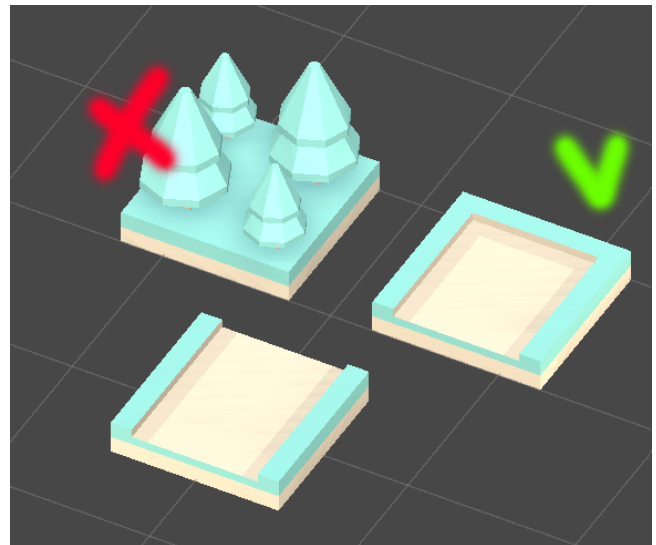


Figure 1: We cannot choose the tile with the trees to be put at the end of the path

In the normal workflow, designers have to design every single mesh and specify their constraints. For example :

- Path : must have a path at the top and bottom, and grass to its left and right
- Dead end : must have grass everywhere except at its bottom
- Trees : must have grass in every direction

The proposed system works in this way :

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '22 Posters, August 07-11, 2022, Vancouver, BC, Canada

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9361-4/22/08.

<https://doi.org/10.1145/3532719.3543198>

The WFC does not choose an initial mesh randomly, it chooses the four constraints of the first cell randomly. Let's say :

- Top : Grass
- Bottom, Left, Right : Path

Then it calls the trained GAN feeding it the 4 constraints, the GAN then gives it back the newly generated mesh that satisfies those constraints, instead of picking one that has been create by an artist.

This is such a mesh :

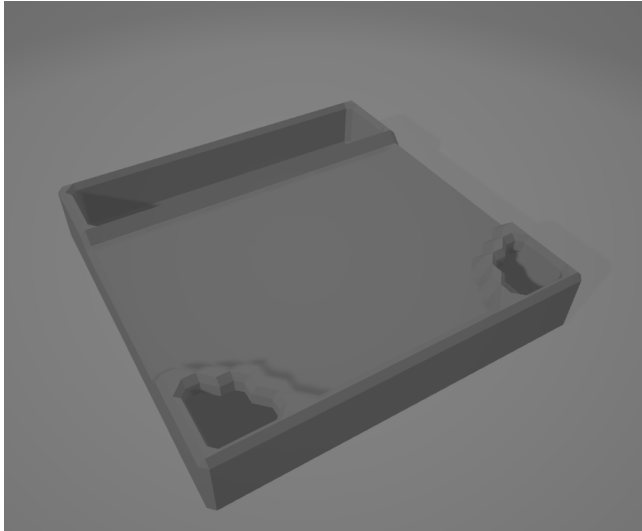


Figure 2: A mesh generated by the GAN which satisfies the constraints : Top = grass, the rest = path

Then for each tile it generates a mesh that satisfies the constraints.

2.2 GAN

The generator aims to produce a voxel grid that is plausible. It has three 3D convolutional layers, each of which includes a bank of $4 \times 4 \times 4$ filters with strides of $2 \times 2 \times 2$, followed by a leaky ReLU activation function. It gives a $32 \times 32 \times 32$ shape result.

The discriminator has the objective of distinguishing whether the estimated 3D shapes are plausible or not. It takes in input both real reconstruction pairs and fake reconstruction pairs. In particular, it consists of three 3D convolutional layers. Each convolutional layer has a bank of $4 \times 4 \times 4$ filters with strides of $2 \times 2 \times 2$, followed by a leaky ReLU activation function except for the last layer which is followed by a sigmoid activation function. The number of output channels of the convolutional layers starts with 64, doubling at each subsequent layer and ends up with 256.

The reconstruction of the mesh is made by using a marching cube function on the voxel grid produced by the generator. For better reconstruction we can use a Poisson reconstruction method but we need a better shape result of at least $128 \times 128 \times 128$.

3 NEXT STEPS

To upgrade to a 3D space. For this we need more meshes which also define constraints in the up and down directions and the associated trained GAN.

4 FUTURE WORKS

In addition to 3D world generation, the proposed system could be used in these ways :

- Environments generation in 3D software like Houdini or Blender
- Using Natural Language Processing AI like Dall-E-2 to interpret a sentence and generate the environment using the proposed system

REFERENCES

- Nordvig Møller, T., Billeskov, J., & Palamas, G. (2020, September). Expanding wave function collapse with growing grids for procedural map generation. In International Conference on the Foundations of Digital Games (pp. 1-4)
- Mintrone, A., & Erioli, A. (2021). Training Spaces-Fostering machine sensibility for spatial assemblages through wave function collapse and reinforcement learning.
- Wang, X., Xu, D., & Gu, F. (2020). 3D model inpainting based on 3D deep convolutional generative adversarial network. *IEEE Access*, 8, 170355-170363.
- Wu, Z., Song, S., Khosla, A., Yu, F., Zhang, L., Tang, X., & Xiao, J. (2015). 3d shapenets: A deep representation for volumetric shapes. In Proceedings of the IEEE conference on computer vision and pattern recognition.
- B. Yang, S. Rosa, A. Markham, N. Trigoni and H. Wen, "Dense 3D Object Reconstruction from a Single Depth View," in *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 12, pp. 2820-2834, 1 Dec. 2019, doi: 10.1109/TPAMI.2018.2868195.